

Who Hunts Who?

Comparing 4 Knowledge Graph Construction Methods for Predator–Prey Relations

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Natural Language Processing

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Motivation & Task

- ▶ Goal: automatically build a **predator-prey knowledge graph** from free text.
- ▶ Source: Wikipedia pages for 18 animals across three trophic levels.
- ▶ No animal-specific hand-written rules — only general NLP signals.
- ▶ Central question: **which extraction approach yields the best trade-off between precision and recall?**

Three trophic levels

Top predators (6)

Hawk, Eagle, Barn Owl, Fox, Wolf, Weasel

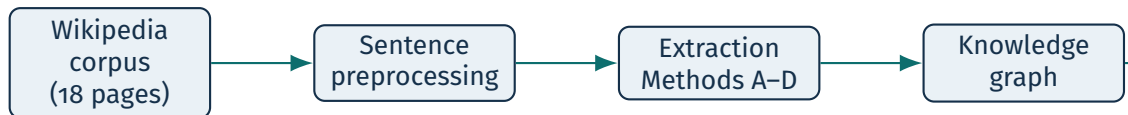
Mid-level (4)

Salamander, Lizard, Spider, Bat

Prey (8)

Mouse, Rabbit, Squirrel, Grasshopper, Cricket, Caterpillar, Fly, Beetle

Pipeline Overview



Corpus

Pages fetched via wikipedia-api, cached as JSON. Text is stripped of markup, parentheticals, and citation markers, then split into sentences.

Gold standard

38 manually curated predator→prey edges. The same benchmark is used for all four methods to ensure a fair comparison.

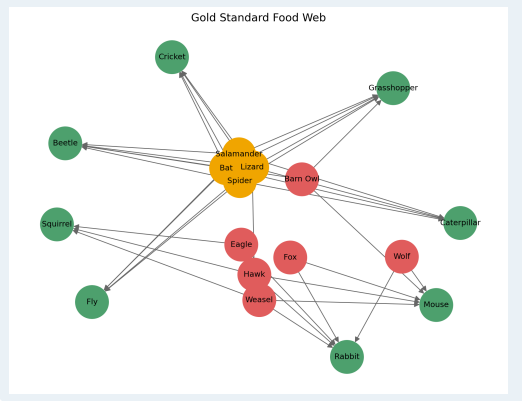
Metrics

Edge-level Precision, Recall, and F1 score. Only edges within valid trophic directions are considered.

Construction rules

- ▶ Manually constructed from Wikipedia text.
- ▶ Directed edges: predator $\rightarrow$ prey.
- ▶ Trophic constraints enforced:
 - top predators may target mid-level *and* prey.
 - mid-level predators may target prey only.
- ▶ 38 edges in total across 10 source animals.

Gold standard graph



Method A — Co-occurrence

How it works

1. For each source animal, iterate over sentences in its Wikipedia page.
2. If any target animal is mentioned in the **same sentence**: add edge source → target.
3. No verb, no syntactic analysis — pure proximity.

Example sentence:

“Foxes and rabbits both live in the forest.”

⇒ adds Fox→Rabbit

even though the sentence is about their *habitat*.

Design trade-offs

Strength: highest possible recall — every real edge that co-occurs in a sentence will be found.

Limitation: no predation signal is required. Any two animals appearing together (as competitors, taxonomic relatives, or examples) become an edge.

Limitation: no direction inference — edges are always assigned as source → target based on which page is being read.

Expected: very high recall, low precision.

Method A vs Gold Standard

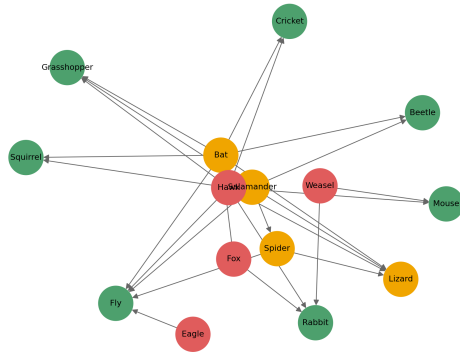
Gold standard

Gold Standard Food Web



Method A

Method A - Co-occurrence



Scores: Precision = 0.346, Recall = 1.000, F1 = 0.514 (TP = 36, FP = 68, FN = 0)

Method B — Predation Verb Patterns

How it works

1. For each sentence containing the source animal, extract a window around both animals.
2. Requires at least one **predation verb or phrase** inside that window:
3. **Infer direction:**
 - Active: “Fox *hunts* Rabbit” → Fox→Rabbit.
 - Passive: “Rabbit *is hunted by* Fox”
Fox→Rabbit.
4. **Category expansion fallback:** “eats insects”, expands to “eats fly”

Design trade-offs

Strength: the verb requirement dramatically reduces false positives compared to Method A.

Limitation: paraphrased predation (“*diet consists of...*”, “*food source for...*”) is missed, lowering recall.

Limitation: direction inference from text position is heuristic; complex sentence structures may be mis-directed.

Limitation: plural forms (mice, flies) not handled uniformly in the window search.

Expected: highest precision, moderate recall.

Method B vs Gold Standard

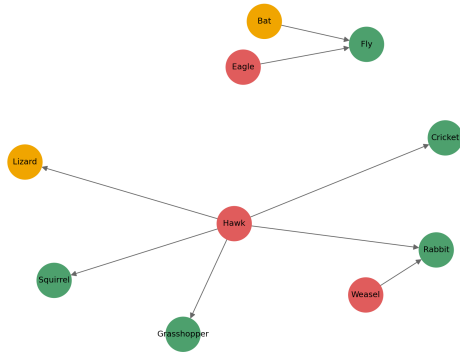
Gold standard

Gold Standard Food Web



Method B

Method B - Dependency Patterns



Scores: Precision = 0.625, Recall = 0.139, F1 = 0.227 (TP = 5, FP = 3, FN = 31)

Method C — OpenIE-style Relation Extraction

Core idea: extract (subject, relation, object) triples.

Step-by-step

1. Build regex from the full predation phrase + keyword vocabulary.
2. Search for `[source] ... [rel] ... [target]` up to 180 characters apart.
3. Handle passive: `[target] ... is eaten by ... [source]`.
4. **Category expansion fallback:** “eats insects”, expands to “eats fly”

Strength: variable-distance regex captures paraphrases missed by Method B’s fixed window.

Strength: category expansion recovers edges when Wikipedia mentions a category (“insects”) rather than a specific animal.

Limitation: generous distance windows may allow subject–relation–object triples that are not actually related.

Limitation: unconstrained category expansion can introduce false positives for all members of a category.

Expected: better recall than B, moderate precision.

Method C vs Gold Standard

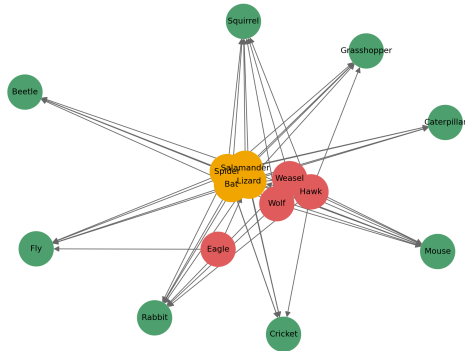
Gold standard

Gold Standard Food Web



Method C

Method C - OpenIE-style Extraction



Scores: Precision = 0.604, Recall = 0.806, F1 = 0.690 (TP = 29, FP = 19, FN = 7)

Method D — Hybrid Extraction (Overview)

Method D combines rule-based extraction with constrained fallback expansion:

▶ **Step 1: Explicit relation extraction**

- Predicate-driven extraction using verb cues + proximity constraints

▶ **Step 2: Restricted category expansion**

- Applied only if Step 1 fails
- Uses insect-based semantic expansion under strict constraints

Key idea: maximize precision first, then recover recall via controlled fallback.

Method D — Step 1: Explicit Relation Extraction

Goal: extract high-precision predator $\rightarrow$ prey edges

Procedure:

- ▶ Identify predation verbs (eat, hunt, kill, consume, attack, etc.)
- ▶ Reject sentences containing negation cues (e.g. not, rarely, never)
- ▶ Detect entity mentions with lexical variants (e.g. mouse/mice)

Edge creation conditions:

- ▶ Both predator and prey appear in the same sentence
- ▶ Both are within a token distance threshold to predation verbs:
 - predator distance $\leq d_{pred}$
 - prey distance $\leq d_{prey}$
- ▶ OR passive construction detected (“X is eaten by Y”)

Effect: high precision extraction with strict linguistic constraints

Method D — Step 2: Restricted Category Expansion

Goal: recover missed edges from Step 1 under controlled conditions

Trigger condition:

- ▶ Only executed if Step 1 does not extract any relation
- ▶ Sentence must contain at least one predation cue

Expansion strategy:

- ▶ Expand prey candidates using insect taxonomy only
- ▶ Apply only to insectivorous source species:
 - Salamander, Lizard, Spider, Bat, Bird

Effect:

- ▶ Improves recall without significantly increasing false positives
- ▶ Acts as a safety-constrained fallback mechanism

Method D — Strengths & Limitations

Strengths

- ▶ Combines explicit linguistic rules with controlled fallback expansion
- ▶ Dual proximity constraint (predator + prey) improves precision
- ▶ Handles passive constructions (e.g. “X is eaten by Y”)
- ▶ Lexical variant matching improves robustness (mouse/mice, etc.)
- ▶ Negation filtering reduces false positive relations

Limitations

- ▶ Relies on manually defined predation vocabulary
- ▶ Token-based distance is only a proxy for syntactic structure
- ▶ Category expansion is restricted to insect prey
- ▶ Cannot capture cross-sentence relations
- ▶ Performance depends on sentence-level parsing quality

Overall: best-performing hybrid trade-off between precision and recall

Method D vs Gold Standard

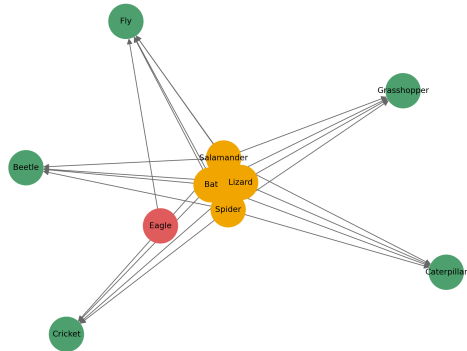
Gold standard

Gold Standard Food Web



Method D

Method D - Hybrid Extraction



Scores: Precision = 0.889, Recall = 0.667, F1 = 0.762 (TP = 24, FP = 3, FN = 12)

Method Design Comparison

	A Co-occurrence	B Verb patterns	C OpenIE	D Hybrid
Predation verb required	No	Yes	Yes	Yes
Direction inference	No	Yes	Yes	Yes
Passive voice handled	—	Yes	Yes	Yes
Category expansion	None	None	Full	Restricted
Proximity constraint	None	Window	Char. distance	Dual token distance
Noisy-context filter	No	No	Yes	Yes
Variant matching	Yes	Partial	Yes	Yes
Expected precision	Low	High	Medium	High
Expected recall	High	Medium	High	Medium-high

Evaluation Metrics

Edge-level evaluation

Let $\hat{E}$ be the set of predicted edges and E^* the gold standard.

$$\text{Precision} = \frac{|\hat{E} \cap E^*|}{|\hat{E}|} \quad \text{Recall} = \frac{|\hat{E} \cap E^*|}{|E^*|}$$

$$F_1 = \frac{2 \cdot P \cdot R}{P + R}$$

- ▶ $|\hat{E} \cap E^*|$ = true positives (TP)
- ▶ $|\hat{E}| - \text{TP}$ = false positives (FP)
- ▶ $|E^*| - \text{TP}$ = false negatives (FN)

Scope constraint

Only edges that satisfy the trophic direction are considered valid:

- ▶ Source must be in **top predators** $\cup$ **mid-level**.
- ▶ Target must be in **mid-level** $\cup$ **prey**.
- ▶ Self-loops are never valid.
- ▶ Top predators cannot appear as prey targets.

All four methods use this same constraint, implemented via `sanitize_edges()`.

Results

Method	P	R	F1	TP	FP	FN
A — Co-occurrence	0.346	0.100	0.514	36	68	0
B — Dependency Patterns	0.625	0.139	0.227	5	3	31
C — OpenIE-style Relation Extraction	0.604	0.806	0.690	29	19	7
D — Hybrid	0.889	0.667	0.762	24	3	12

Error Analysis

▶ Method A — Co-occurrence

- High recall due to saturation strategy
- Very large number of false positives (semantic noise)
- Lacks directional and relational constraints

▶ Method B — Dependency Patterns

- Very low recall due to strict pattern dependency
- Fails on lexical variation and non-canonical syntax
- High precision but extremely sparse predictions

▶ Method C — OpenIE-style

- Best recall (0.806) but introduces structural noise
- Errors from overly permissive relation scopes
- Sensitive to ambiguous or generic relation phrases

▶ Method D — Hybrid

- Strong precision (0.889), low FP rate
- Remaining errors mainly due to:
 - ▶ long-distance dependencies beyond threshold
 - ▶ missed implicit relations without explicit cue

Discussion

- ▶ **Key finding: trade-off between precision and recall**
 - Rule-based constraints significantly improve precision
 - However, stricter filtering reduces recall
- ▶ **Best overall method: Hybrid (D)**
 - Achieves best F1 (0.762) across all methods
 - Balances explicit linguistic rules with controlled fallback expansion
 - Strong reduction in false positives compared to C
- ▶ **Main bottleneck**
 - Recall limited by proximity thresholds and sentence-level extraction
 - Cross-sentence and long-range dependencies remain unhandled
- ▶ **Interpretation**
 - A: recall-heavy baseline
 - B: precision-heavy but brittle system
 - C: recall-maximising noisy extractor
 - D: controlled balance between both extremes

Summary

- **Conclusion:** Structured rule constraints improve precision, but recall depends on how strictly relational evidence is enforced

	Method	Core signal	Observed behaviour
A	Co-occurrence	sentence co-occurrence	high recall, low precision (no)
B	Verb patterns	strict predation verb + window constraint	very low recall, brittle extraction
C	OpenIE-style	relation patterns + category expansion	highest recall, moderate noise
D	Hybrid	verb + dual proximity + restricted fallback	best F1, strongest precision-recall